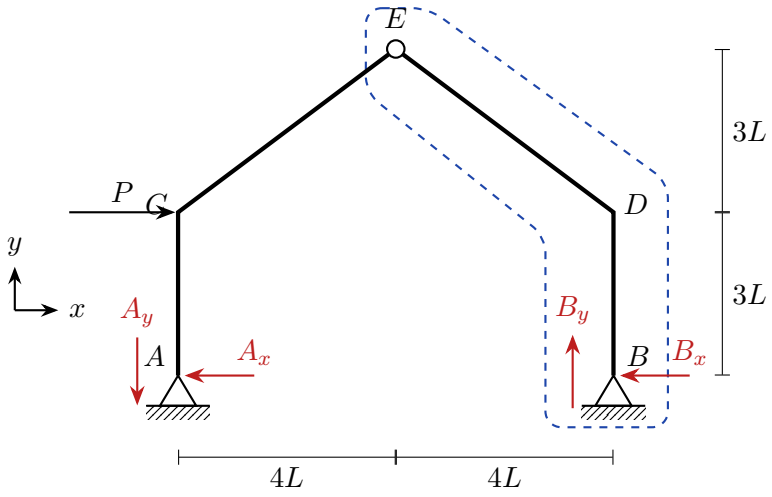
Universidad Técnica Federico Santa María
Departamento de Obras Civiles
CIV-233 Fundamentos de Análisis Estructural

Pauta Certamen 1

Problema 1 (40 pts)



Reacciones

$$\sum M_B = 0 \;\Rightarrow\; P \cdot 3L = A_y \cdot 8L \;\Rightarrow\; \boxed{A_y = \tfrac{3}{8}P} \quad (2)$$

$$\sum F_y = 0 \;\Rightarrow\; B_y = A_y \;\Rightarrow\; \boxed{B_y = \tfrac{3}{8}P} \quad (2)$$

$$\sum F_x = 0 \;\Rightarrow\; A_x + B_x = P$$

Equilibrio de la parte derecha (rótula en E, zona encerrada en línea segmentada):

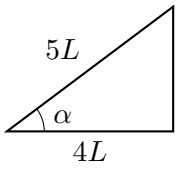
$$\sum M_E^{\text{der}} = 0 \;\Rightarrow\; B_x \cdot 6L = B_y \cdot 4L \;\Rightarrow\; B_x = \frac{3}{8}P \cdot \frac{4L}{6L}$$

$$\Rightarrow\; \boxed{B_x = \frac{P}{4}} \quad (2)$$

$$\Rightarrow\; \boxed{A_x = \frac{3P}{4}} \quad (2)$$

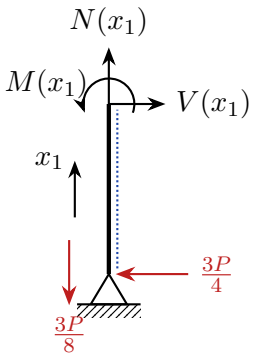
Diagramas de fuerzas internas

Geometría de las vigas inclinadas:



$$\cos\alpha = \frac{4}{5}\,, \qquad \sin\alpha = \frac{3}{5}$$

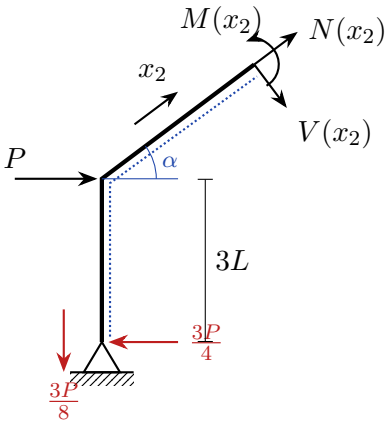
- **Columna** AC $\quad (0 \le x_1 \le 3L)$



$$\sum M_0 = 0 \;\Rightarrow\; \boxed{M(x_1) = \frac{3}{4} P\, x_1} \qquad (5)$$

$$M(x_1 = 0) = 0\,, \qquad M(x_1 = 3L) = \frac{9PL}{4}$$

- **Viga** CE $\quad (0 \le x_2 \le 5L)$



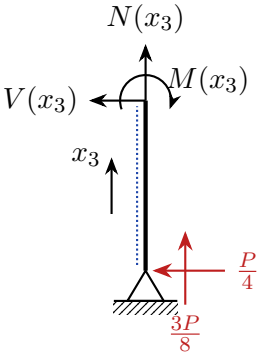
$$\sum M_0 = 0 \;\Rightarrow\; M(x_2) = \frac{3P}{4}\,(3L + x_2 \sin\alpha) - \frac{3P}{8}\, x_2 \cos\alpha - P\, x_2 \sin\alpha$$

$$M(x_2) = \frac{9PL}{4} + \frac{3P}{4} \cdot \frac{3}{5} x_2 - \frac{3}{8} P \cdot \frac{4}{5} x_2 - P \cdot \frac{3}{5} x_2$$

$$\boxed{M(x_2) = -\frac{9}{20} P\,(x_2 - 5L)} \qquad (5)$$

$$M(x_2 = 0) = \frac{9PL}{4}\,, \qquad M(x_2 = 5L) = 0$$

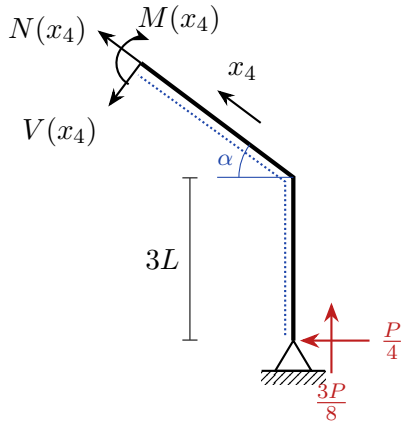
- **Columna** BD $\quad (0 \le x_3 \le 3L)$



$$\boxed{M(x_3) = -\frac{P}{4}\, x_3} \qquad (5)$$

$$M(x_3 = 0) = 0\,, \qquad M(x_3 = 3L) = -\frac{3PL}{4}$$

- **Viga** $DE \quad (0 \le x_4 \le 5L)$



$$\sum M_0 = 0 \;\Rightarrow\; M(x_4) = \frac{3}{8}P\,x_4\cos\alpha - \frac{P}{4}\,(x_4\sin\alpha + 3L)$$

$$M(x_4) = \frac{3}{8}P\cdot\frac{4}{5}x_4 - \frac{P}{4}\cdot\frac{3}{5}x_4 - P\cdot\frac{3L}{4}$$

$$M(x_4) = \frac{3}{20}P\,x_4 - \frac{3PL}{4}$$

$$\boxed{M(x_4) = \frac{3}{20}P\,(x_4 - 5L)} \qquad (5)$$

$$M(x_4 = 0) = -\frac{3PL}{4}, \qquad M(x_4 = 5L) = 0$$

Cálculo del desplazamiento $\Delta_{C,x}$

Teorema de Clapeyron: $W = U$

$$\boxed{W = \frac{1}{2}P\,\Delta_{C,x}} \qquad (2)$$

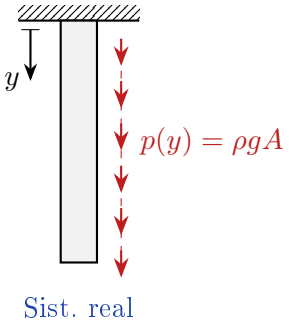
$$U = \frac{1}{2}\int_{L_T}\frac{M(x)^2}{EI}\,dx$$

$$U = \frac{1}{2}\int_0^{3L}\frac{\left(\frac{3}{4}Px_1\right)^2}{EI}\,dx_1 + \frac{1}{2}\int_0^{5L}\frac{\left(-\frac{9}{20}P\right)^2(x_2-5L)^2}{EI}\,dx_2$$
$$+\frac{1}{2}\int_0^{3L}\frac{\left(-\frac{P}{4}x_3\right)^2}{EI}\,dx_3 + \frac{1}{2}\int_0^{5L}\frac{\left(\frac{3}{20}P\right)^2(x_4-5L)^2}{EI}\,dx_4$$

$$\boxed{U = \frac{1}{2EI}\left(\frac{9}{16}P^2\cdot\frac{27L^3}{3} + \frac{81}{400}P^2\cdot\frac{125L^3}{3} + \frac{P^2}{16}\cdot\frac{27L^3}{3} + \frac{9}{400}P^2\cdot\frac{125L^3}{3}\right)} \qquad (8)$$

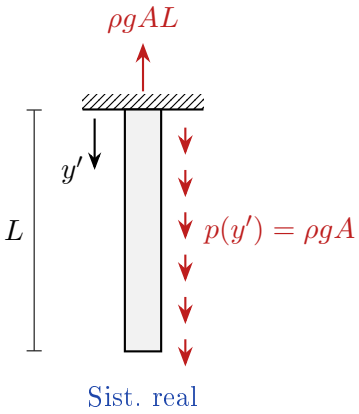
$$U = \frac{15\,P^2L^3}{2EI} \;\Rightarrow\; \boxed{\Delta_{C,x} = \frac{15\,PL^3}{EI}} \qquad (2)$$

Problema 2 (60 pts)

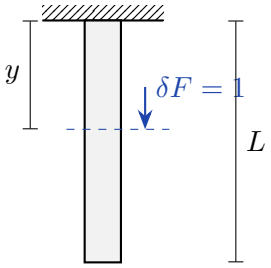


Sist. real

a) Principio de las Fuerzas Virtuales (PFV)

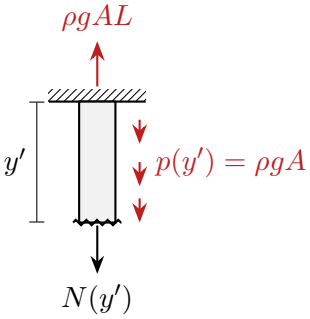


Sist. real



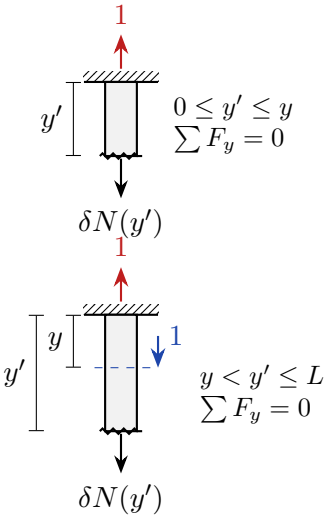
Sist. virtual (5)

- **Estática sistema real**



$$\sum F_y = 0 \;\Rightarrow\; N(y') = \rho g A\,(L - y') \quad (5)$$

- **Estática sistema virtual**



$$\delta N(y') = 1, \quad 0 \le y' \le y$$
$$\delta N(y') = 0, \quad y < y' \le L \quad (5)$$

Aplicando el PFV ($\delta W^* = \delta U^*$), con $\varepsilon_{yy} = \sigma_{yy}/E$:

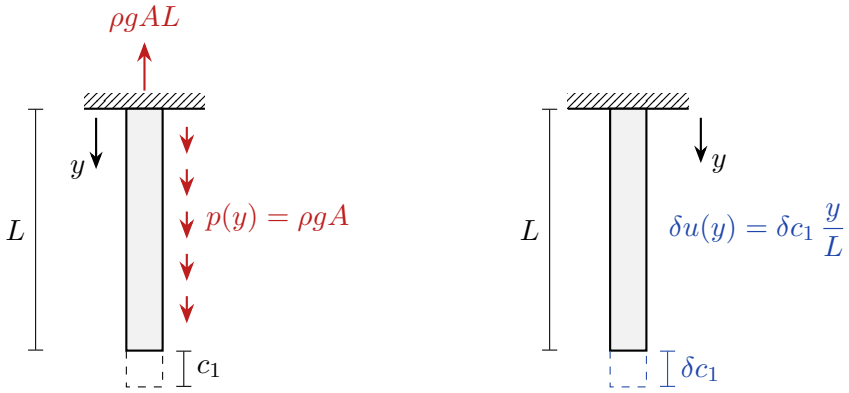
$$1 \cdot u(y) = \int_V \delta\sigma_{yy}\,\varepsilon_{yy}\,dV \qquad (5)$$

$$u(y) = \int_V \frac{\delta N(y')}{A} \cdot \frac{N(y')}{EA}\,dV$$

$$u(y) = \int_0^y A \cdot \frac{1}{A} \cdot \frac{\rho g A\,(L-y')}{EA}\,dy' + \underbrace{\int_y^L A \cdot \frac{0}{A} \cdot \frac{\rho g A\,(L-y')}{EA}\,dy'}_{=0}$$

$$u(y) = \frac{\rho g}{E}\left(Ly - \frac{y^2}{2}\right) \qquad (5)$$

b) Principio de los Desplazamientos Virtuales (PDV)



Sist. virtual (5)

Sea el campo de desplazamientos

$$u(y) = c_1\,\frac{y}{L}\,, \qquad \text{como } u(y=L) = c_1\,,$$

c_1 corresponde al desplazamiento vertical del extremo de la barra. (5)

Trabajo virtual externo:

$$\delta W = \int_0^L p(y)\,\delta u(y)\,dy = \int_0^L \rho g A\,\delta c_1\,\frac{y}{L}\,dy$$

$$\delta W = \frac{\rho g A L\,\delta c_1}{2} \qquad (5)$$

Trabajo virtual interno: con $\sigma_{yy} = E\,\varepsilon_{yy}$,

$$\delta U = \int_V \sigma_{yy}\,\delta\varepsilon_{yy}\,dV$$

$$\varepsilon_{yy} = \frac{d}{dy}u(y) = \frac{c_1}{L}\,, \qquad \delta\varepsilon_{yy} = \frac{d}{dy}\delta u(y) = \frac{\delta c_1}{L}$$

$$\delta U = \int_0^L A\,E\,\frac{c_1}{L}\,\frac{\delta c_1}{L}\,dy \;\Rightarrow\; \boxed{\delta U = \frac{AE\,c_1\,\delta c_1}{L}} \qquad (5)$$

$$\delta W = \delta U \;\Rightarrow\; \boxed{c_1 = \frac{\rho g L^2}{2E}} \qquad (5)$$

De modo que:

$$u(y) = \frac{\rho g L}{2E}\,y$$

c) Solución exacta (ecuación diferencial de equilibrio)

$$\frac{\partial \sigma_{yy}}{\partial y} + f_y = 0\,, \qquad f_y = \frac{p(y)}{A}$$

$$\frac{\partial \sigma_{yy}}{\partial y} = -f_y = -\rho g \quad \Big/\int$$

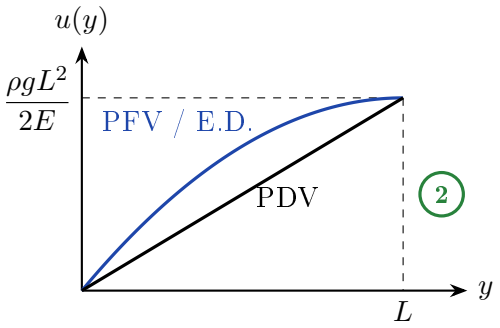
$$\boxed{\sigma_{yy}(y) = -\rho g\,y + C_1} \quad (2)\,; \qquad \sigma_{yy}(y=0) = \frac{\rho g A L}{A} \;\Rightarrow\; C_1 = \rho g L$$

$$\varepsilon_{yy} = \frac{\sigma_{yy}}{E} \;\Rightarrow\; \boxed{\varepsilon_{yy} = \frac{-\rho g\,y + \rho g L}{E}} \qquad (2)$$

$$\frac{du(y)}{dy} = \varepsilon_{yy} \quad \Big/\int \;\Rightarrow\; u(y) = -\frac{\rho g\,y^2}{2E} + \frac{\rho g L\,y}{E} + C_2$$

$$u(y=0) = 0 \;\Rightarrow\; C_2 = 0$$

$$\therefore \quad \boxed{u(y) = \frac{\rho g}{E}\left(Ly - \frac{y^2}{2}\right)} \qquad (2)$$



(2)

El campo de desplazamientos correcto es el determinado por PFV, que corresponde al mismo determinado por las ecuaciones diferenciales. El campo de desplazamientos determinado por PDV sólo es una aproximación lineal.

(2)